

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0368

Brooklyn City North Plant 138 kV
August 17, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0368. The Interconnection Customer proposes a 169.6 MW wind facility interconnecting at the Brooklyn City North Plant 138 kV substation in the Virginia area. The request was received on August 25, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 163 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,179,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
International Paper Valliant Ok Substation to Brooklyn City N Plant 138.0 kV	American to Urban 345.0 kV	125.1	204	71.7
Brightn to G-P Gypsum 138.0 kV	Artesia Country Club to Air Force Base 345.0 kV	108.9	331	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
International Paper Valliant Ok Substation to Brooklyn City N Plant 138.0 kV	Rebuild line with higher-capacity conductor	\$1,556,000	89% (\$1,386,000)
G-P Gypsum to Brighton Substation 138.0 kV	Reconductor line	\$2,265,000	35% (\$793,000)
	Total allocated to project		\$2,179,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.